

BIOMECHANICS (EPHE 341)

Forces

Optional reading: McGinnis Chpt 1 pg 20–
32, 39

Force

Biomechanics is the study of the effects of internal and external **forces** on the human body in movement and rest.

Simply defined a force is a **push or a pull**.

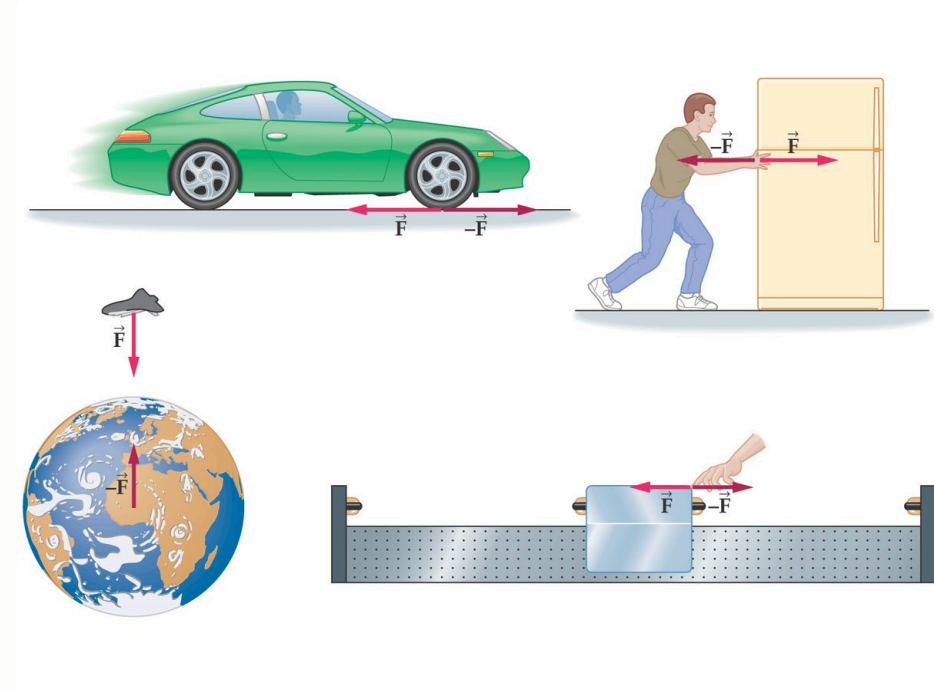
Force is a **vector** quantity

Can cause a change in the state of motion of an object.

start, stop, speed up, slow down or change direction

Can either accelerate and/or deform an object

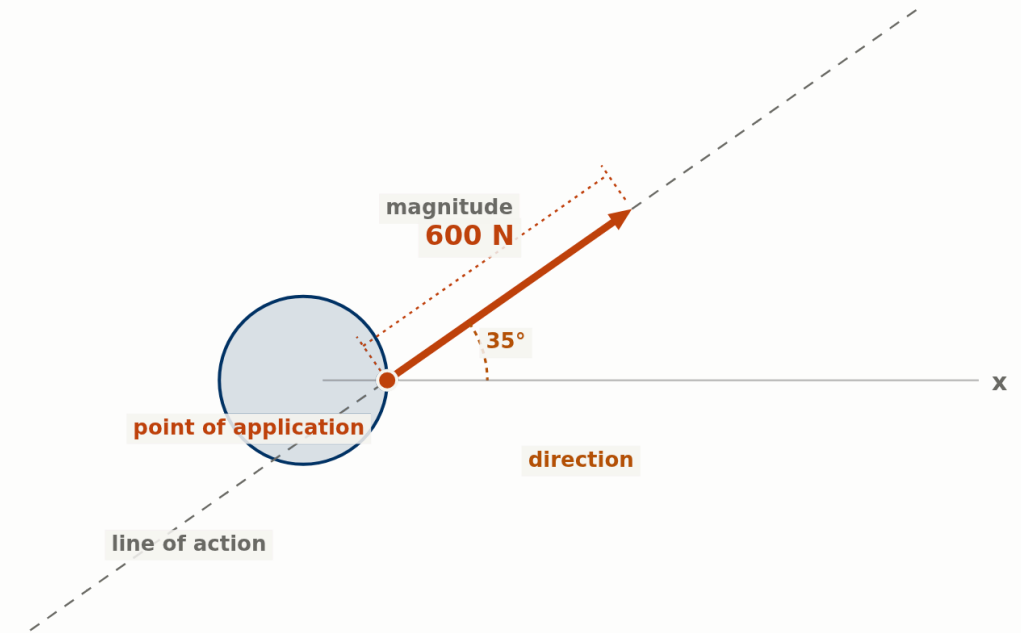
comes in pairs: action and reaction



Force vectors

Some important points about force vectors

- **Magnitude** – length of the arrow
- **Direction** – relative to an axis
- **Line of action** – the shaft of the arrow
- **Point of application** – where it acts on an object



600 N at **35°** above the x axis · applied at the **centre**

Magnitude changes the length, direction swings the whole line of action, and the point of application decides where on the object it acts. Move it off the centre.

through the centre

at the top edge

at the side

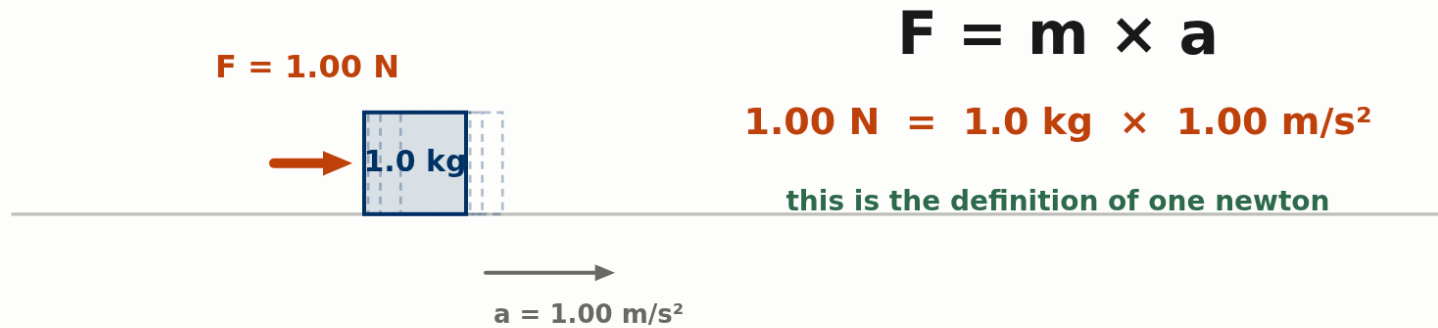
Magnitude 600 N

Direction 35°

Force — units

The Newton (N) is the SI unit of measurement for force. One N of force is defined as: the force required to accelerate a 1 kg mass at 1 m/s². $1 \text{ N} = (1.0 \text{ kg})(1.0 \text{ m/s}^2)$

The equation for force (F) is $\mathbf{F = ma}$, where **m** is mass and **a** is acceleration.



$$1.00 \text{ N} = 1.0 \text{ kg} \times 1.00 \text{ m/s}^2$$

One newton is the force that accelerates one kilogram at one metre per second squared — set both sliders to 1 and the equation says so. Doubling either one doubles the force.

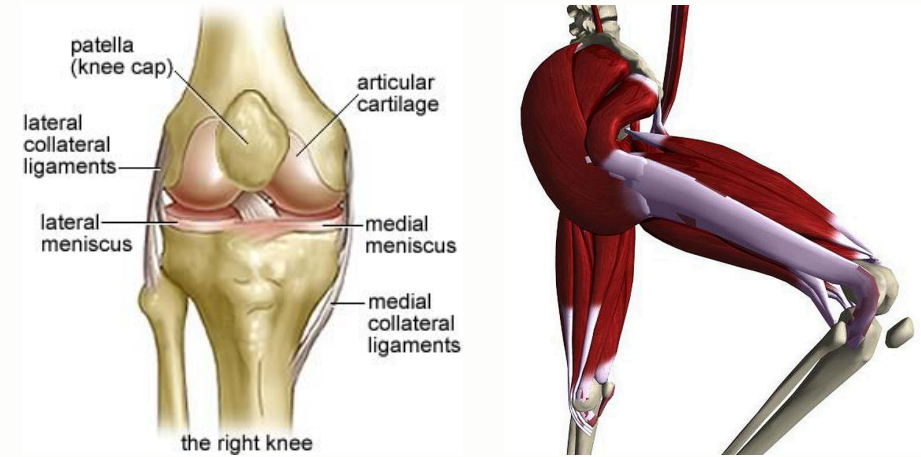
Mass	<input type="range" value="1"/>	1.0 kg
Acceleration	<input type="range" value="1"/>	1.00 m/s ²

Internal forces

Internal forces act within an object or system.

Compressive (pushing) and **tensile** (pulling) forces.

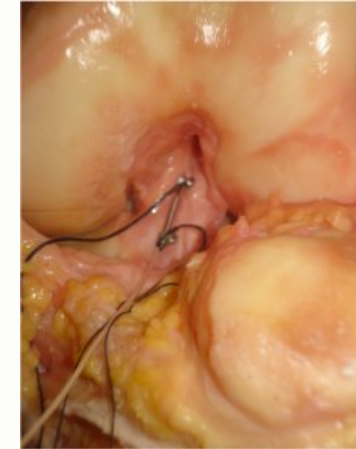
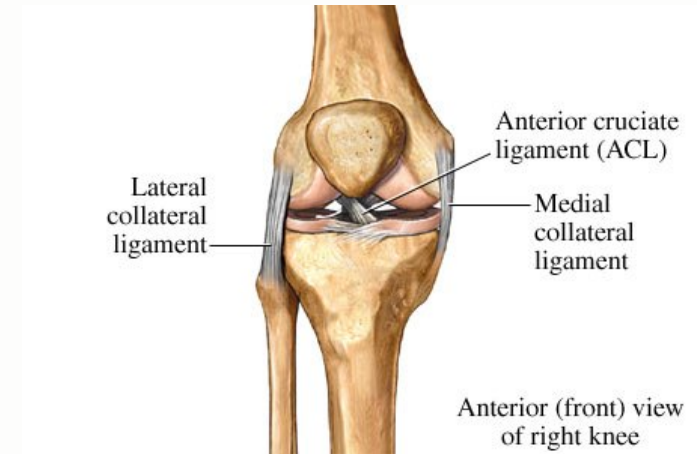
When you land from a jump there is a force from your lower leg on your upper leg through your knee and an equal and opposite force from your upper leg to your lower leg through your knee.



Measuring internal forces in biomechanics

It is difficult to directly measure internal forces in human biomechanics.

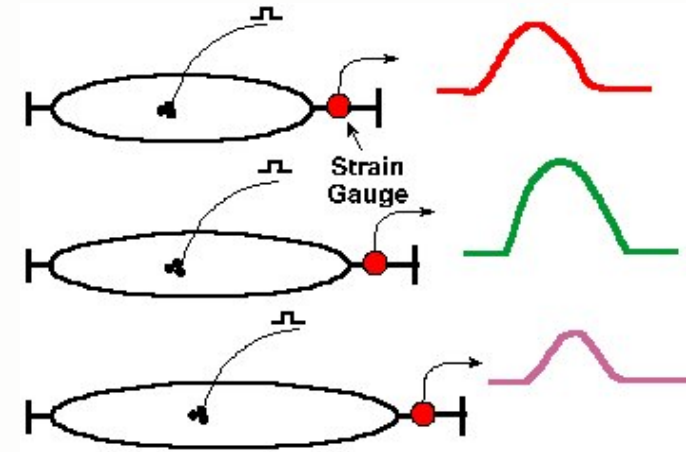
An invasive option is to use **strain gauges in cadaver limbs** to conduct testing of injury occurrence and prevention.



Measuring internal forces in biomechanics

Strain gauge or optical fiber on/in human or animal tendon can be used to measure the **magnitude** of muscle forces.

But what about the **direction**?



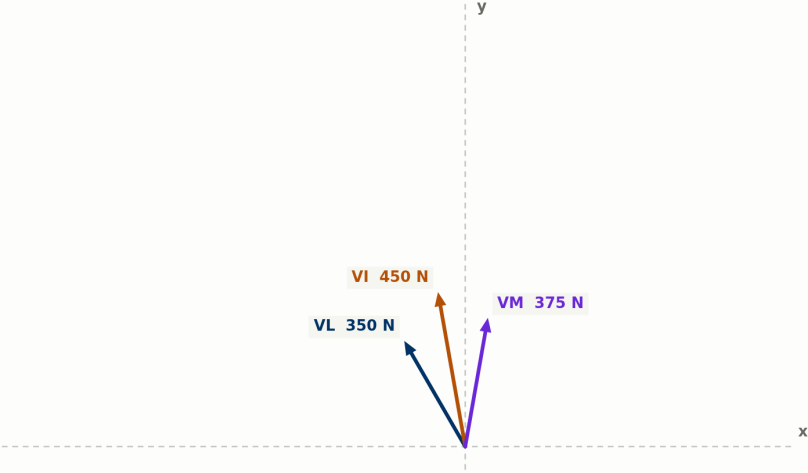
Step 1: Determine what the question is asking for · Step 2: Draw a diagram with all known values

Using fibre optic force measurement we have determined the muscle forces of the quadriceps during a squat. Find the resultant quadriceps muscle force given the individual muscle forces of vastus lateralis (VL), vastus intermedialis (VI) and vastus medialis (VM)

VL = 350 N 60°

VI = 450 N 80°

VM = 375 N 80°



Step 2 · draw what is known

	force	angle
VL	350 N	60°
VI	450 N	80°
VM	375 N	80°

The angle is measured from the horizontal; VL and VI pull to one side of the midline, VM to the other.

R = **1131.3 N** at **99.6°** · $R_x = -188.0\text{ N}$, $R_y = 1115.6\text{ N}$

Three forces of 350, 450 and 375 N add to 1131 N, not 1175 N — the sideways pulls partly cancel.

1 · the question

2 · the diagram

3 · components

4 · add them

5 · the resultant

VL

VI

VM

Back to the lecture's numbers

Force

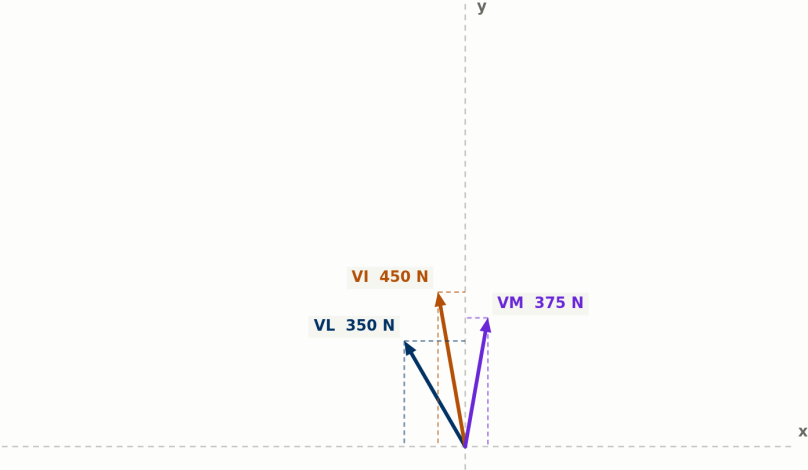
350 N

Internal force vector resolution

Step 3: Isolate the forces and determine force components using trig functions

It is easiest to isolate each muscle force and then determine the separate components.

	HORIZONTAL	VERTICAL
VL 350 N at 60°	−175 N	303 N
VI 450 N at 80°	−78.1 N	443 N
VM 375 N at 80°	65.1 N	369 N



Step 3 · components of each force

VL — vastus lateralis

$$F_x = F \cdot \cos\theta = 350 \cdot \cos 120^\circ = \textcolor{red}{-175.0} \text{ N}$$

$$F_y = F \cdot \sin\theta = 350 \cdot \sin 120^\circ = \textcolor{red}{303.1} \text{ N}$$

	F_x	F_y
VL	−175.0	303.1
VI	−78.1	443.2
VM	65.1	369.3

$$R = \textcolor{red}{1131.3} \text{ N at } \textcolor{red}{99.6}^\circ \cdot R_x = -188.0 \text{ N}, R_y = 1115.6 \text{ N}$$

Three forces of 350, 450 and 375 N add to 1131 N, not 1175 N — the sideways pulls partly cancel.

1 · the question

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VL

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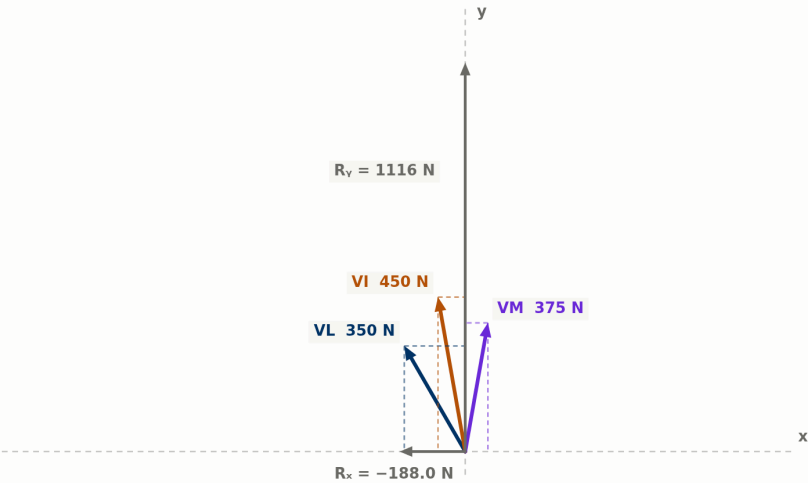
Back to the lecture's numbers

Step 4: Add the respective force components to determine the resultant vector component

You could simply add the forces or use a table similar to when adding linear kinematic variables

$$R_x = \Sigma \text{ Horizontal Forces}$$
$$= -175 \text{ N} - 78.1 \text{ N} + 65.1 \text{ N} = \textbf{-188.8 N}$$

$$R_y = \Sigma \text{ Vertical Forces}$$
$$= 303 \text{ N} + 443 \text{ N} + 369 \text{ N} = \textbf{1115 N}$$



Step 4 · add the components

	F _x	F _y	Horizontal with horizontal, vertical with vertical — the same table you used for kinematics.
VL	-175.0	303.1	
VI	-78.1	443.2	
VM	65.1	369.3	
Σ	-188.0	1115.6	

R = **1131.3 N** at **99.6°** · R_x = -188.0 N, R_y = 1115.6 N

Three forces of 350, 450 and 375 N add to 1131 N, not 1175 N — the sideways pulls partly cancel.

1 · the question

2 · the diagram

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VL

VI

VM

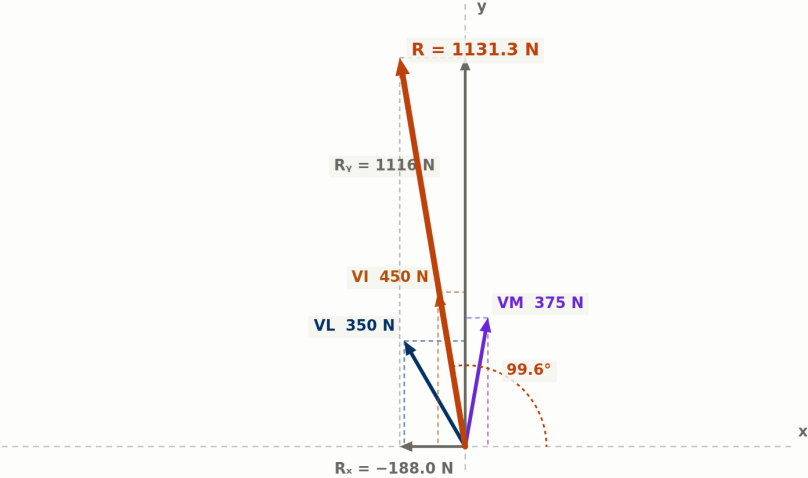
Back to the lecture's numbers

Force 350 N

Step 5: Solve for the resultant force magnitude and direction

$$R^2 = R_x^2 + R_y^2$$
$$= -188.8^2 + 1115^2$$
$$R = \mathbf{1130.87\text{ N}}$$

$$\tan \theta = R_y/R_x$$
$$\theta = \tan^{-1} (R_y/R_x)$$
$$\theta = \tan^{-1} (1115 / -188.8)$$
$$= \mathbf{-80.4}$$



Step 5 · magnitude and direction

magnitude

$$R^2 = R_x^2 + R_y^2$$
$$R = \mathbf{1131.3\text{ N}}$$

direction

$$\tan \theta = R_y / R_x = 1116 / -188.0$$
$$\theta = \mathbf{99.6^\circ}$$
 from the x axis

Watch the quadrant. A calculator given 1116 / -188.0 returns -80.4°, which points down and to the right. Rx is negative and Ry is positive, so the resultant is up and to the left: add 180° to get 99.6°.

R = **1131.3 N** at **99.6°** · Rx = -188.0 N, Ry = 1115.6 N

Three forces of 350, 450 and 375 N add to 1131 N, not 1175 N — the sideways pulls partly cancel.

- 1 · the question
- 2 · the diagram
- 3 · components
- 4 · add them

5 · the resultant

VL

VI

VM

Back to the lecture's numbers

Force 350 N

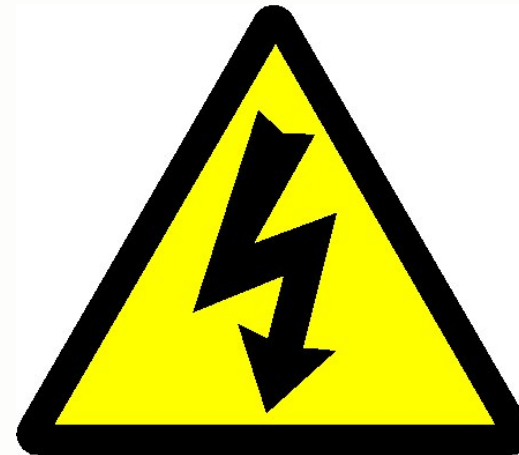
External forces

Forces that act on an object as a result of the surroundings.

Contact and non-contact.

Non-contact forces

- Gravity
- Magnetic
- electrical



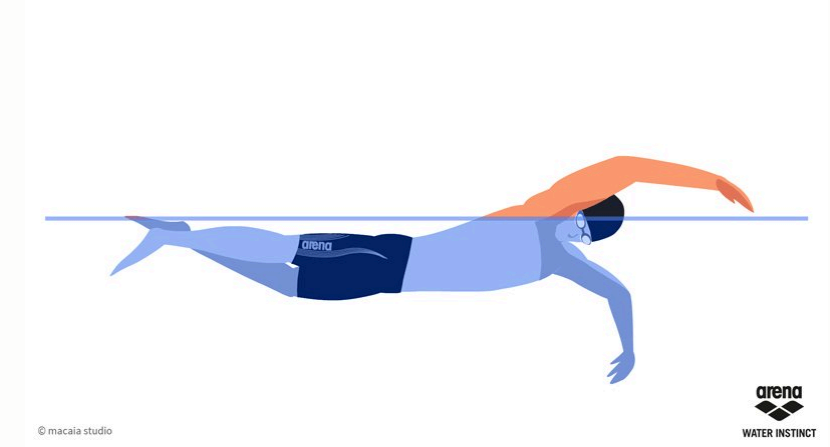
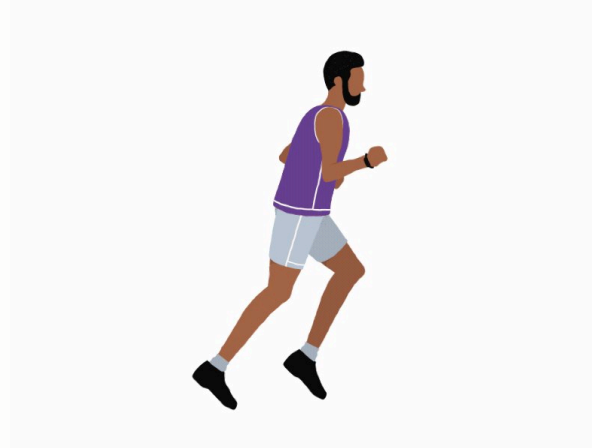
Gravity and contact forces

Contact forces

Occur between objects in contact with each other — solid or fluid. Air resistance and water resistance are fluid contact forces.

Occur in pairs: **action and reaction**.

The force of gravity acting on us causes us to contact the ground.

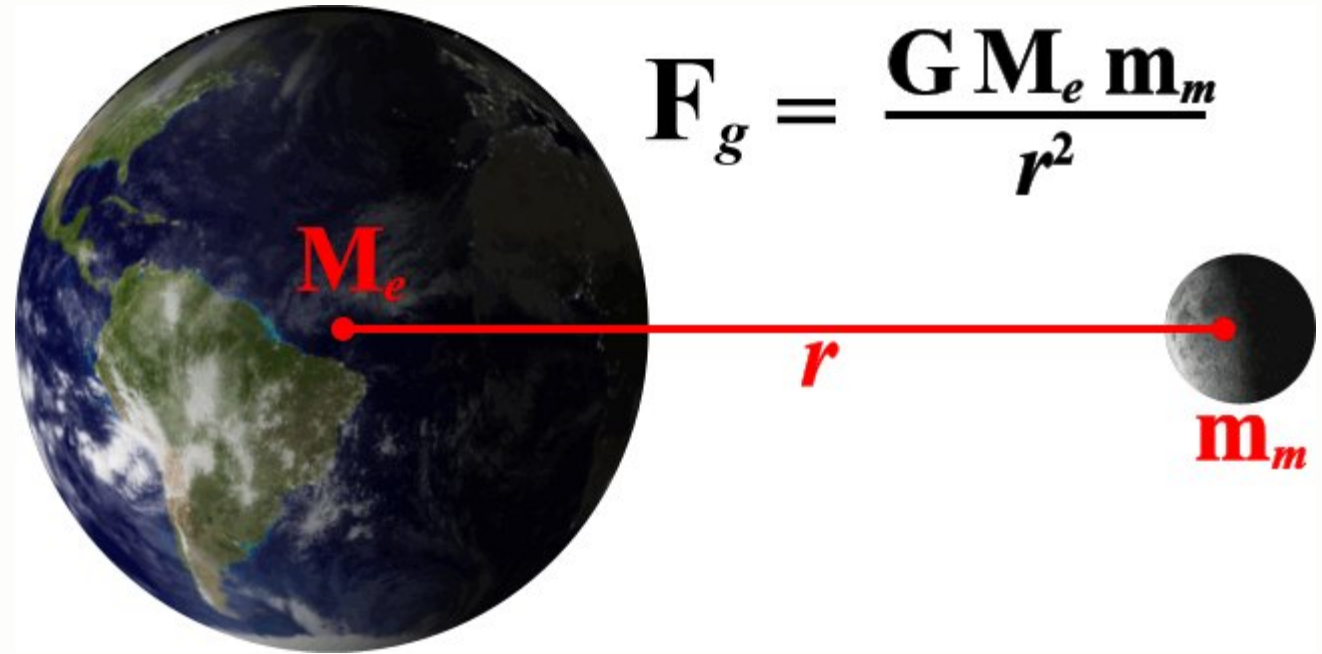


Gravity

We experience gravity as **weight**.

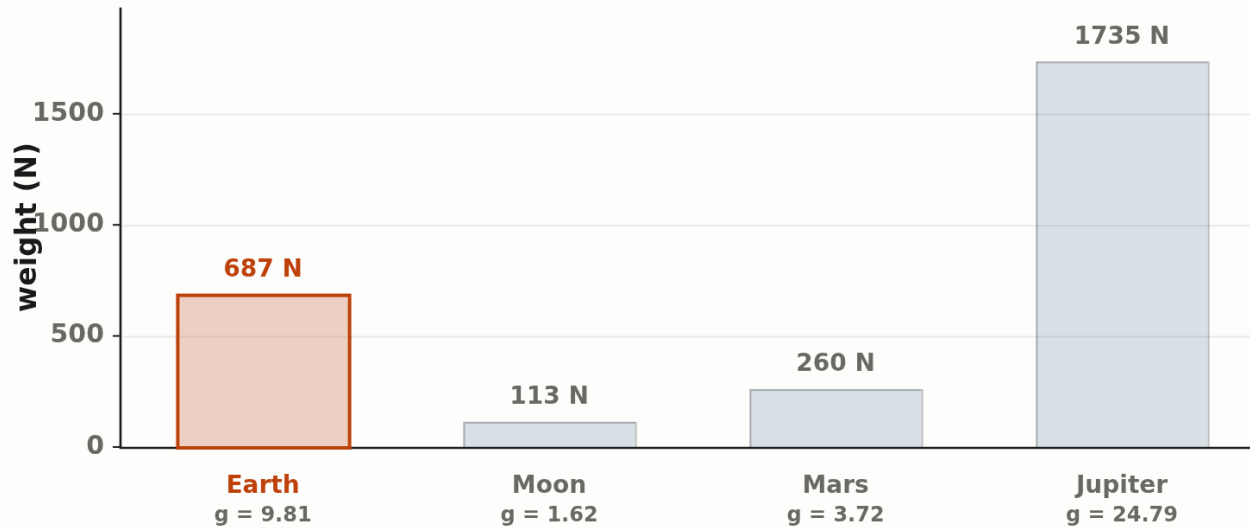
All objects with mass have an attraction to other objects with mass. The magnitude of the attraction depends on the mass of each object and the distance between the two objects. The earth attracts someone and the person attracts the earth with an equal but opposite force.

We have a constant force acting on us pointed towards the centre of the earth acting through our **centre of mass**.



Gravity and weight

The weight of an object is defined as the force of gravity acting on an object. We know that the acceleration due to gravity is 9.81 m/s^2 , and the equation for force is $F = ma$.



$W = mg = 70 \times 9.81 = 687 \text{ N}$ · on Earth

The mass never changes — 70 kg is 70 kg anywhere. The weight is a **force**, and it changes with whatever is pulling on you. On Earth, multiply kilograms by about ten to get newtons.

Body mass 70 kg

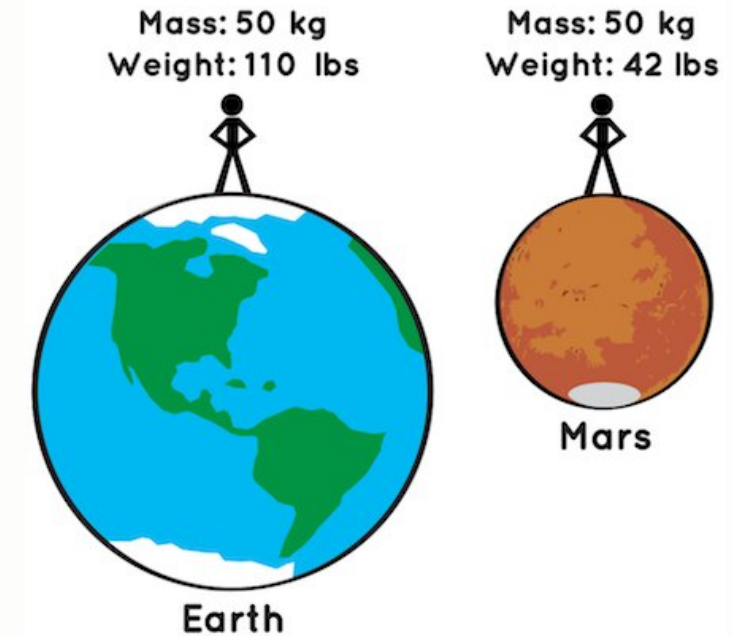
Earth

Moon

Mars

Jupiter

Therefore your weight (W) is equal to your body mass in kilograms (m) multiplied by the acceleration due to gravity ($g = 9.81 \text{ m/s}^2$), $W = mg$. **What is your weight in Newtons?**



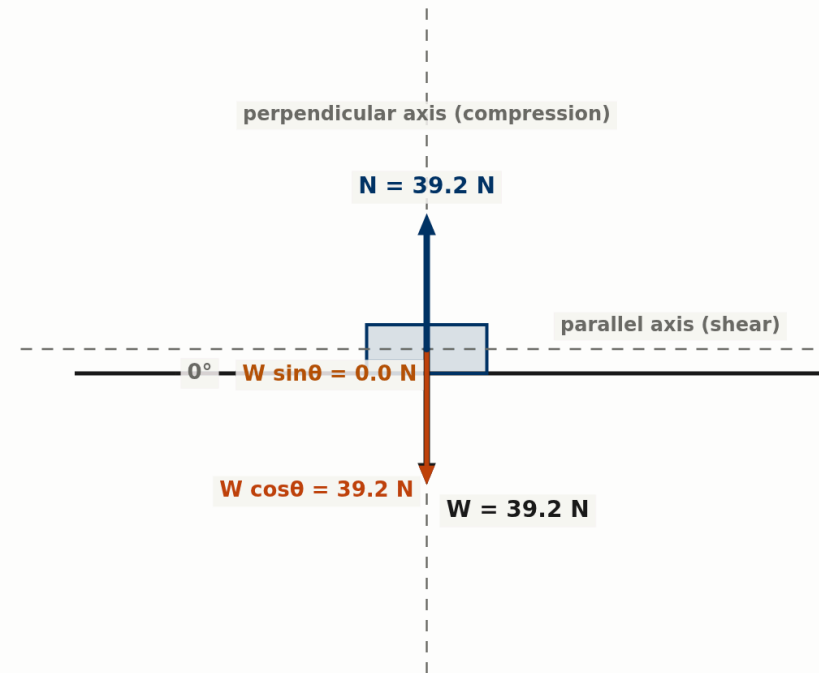
Force of gravity and normal force

The force of gravity is acting towards the earth through our center.

This causes us to contact the ground and the ground must exert a force to hold us up just as we must exert muscular force to remain upright.

The component of the force of the ground acting on us perpendicular to the contact surface is called the **normal force**.

When we are on a surface parallel to the horizontal, the normal force is equal in magnitude to the force of gravity and opposite in direction....



Flat ground: the normal force is **equal and opposite** to the weight, 39.2 N.

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle 0°

Mass 4 kg

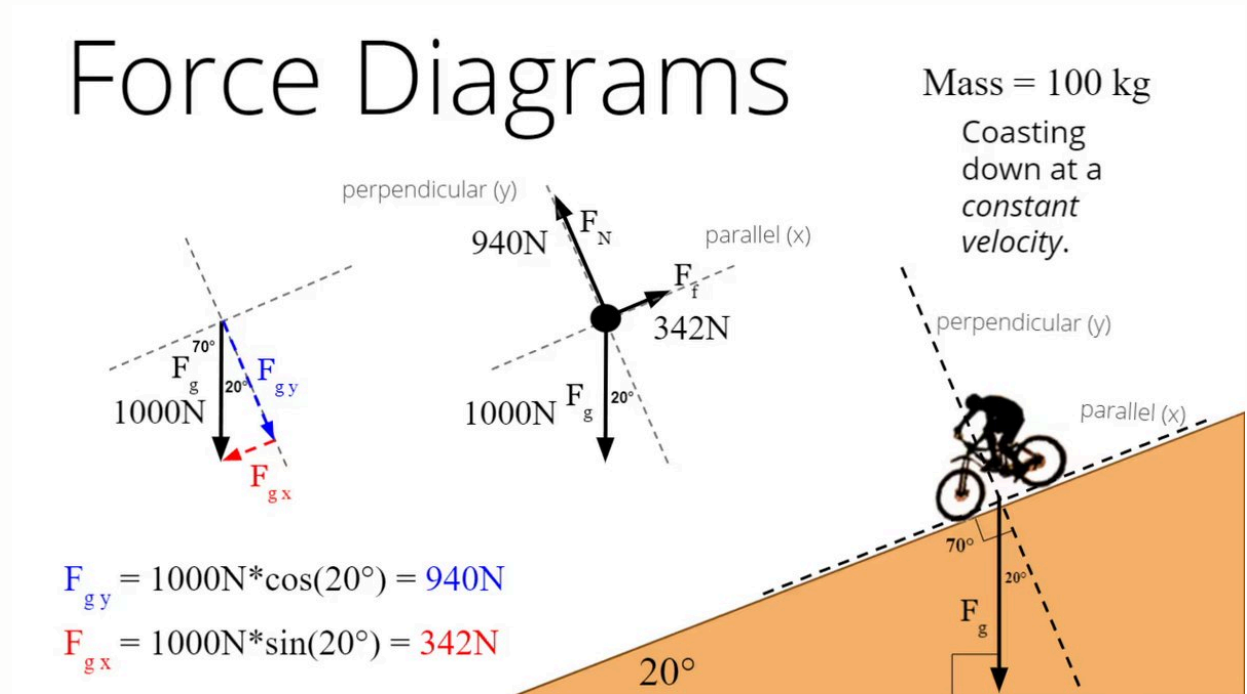
flat ground 4 kg at 20° a 102 kg cyclist at 20°

Force of gravity and normal force

When we or an object are on a **slanted surface**:

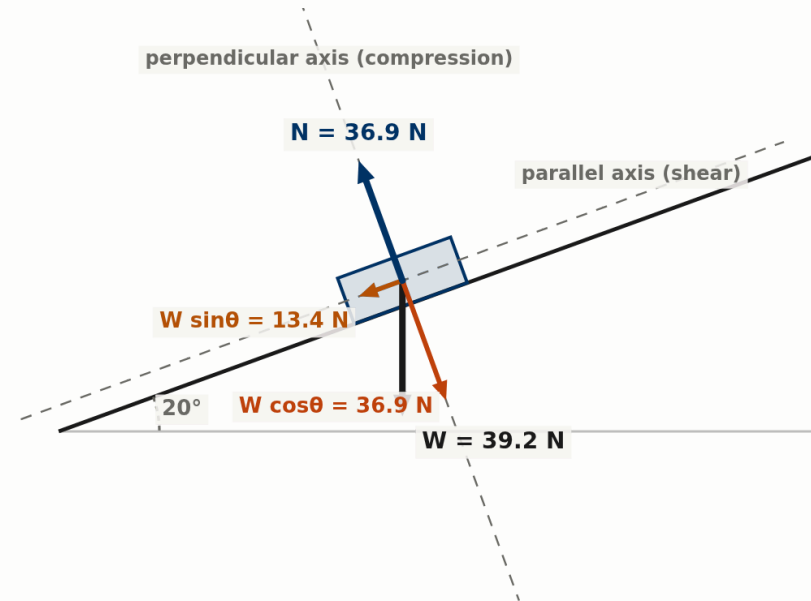
The force of gravity vector still points directly towards the centre of the earth but the normal force stays perpendicular to the contact surface and is no longer equal and opposite to the magnitude and direction of the force of gravity.

So how do we calculate the normal force?



Force

With contact forces it is customary to create an axis system that is **parallel and perpendicular to the contact surface**.



At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle 20°

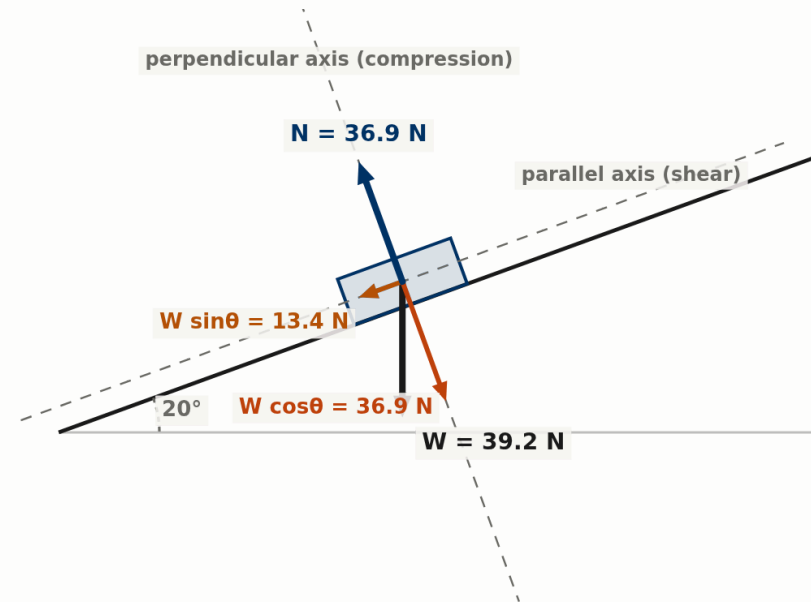
Mass 4 kg

flat ground 4 kg at 20° a 102 kg cyclist at 20°

Then any forces acting in the system can be resolved into components that are:

- Perpendicular to the contact surface (**compression forces**) and
- Parallel to the contact surface (**shear forces**).

The normal force is equal and opposite to the component of gravity force that is perpendicular to the contact surface.



At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle 20°

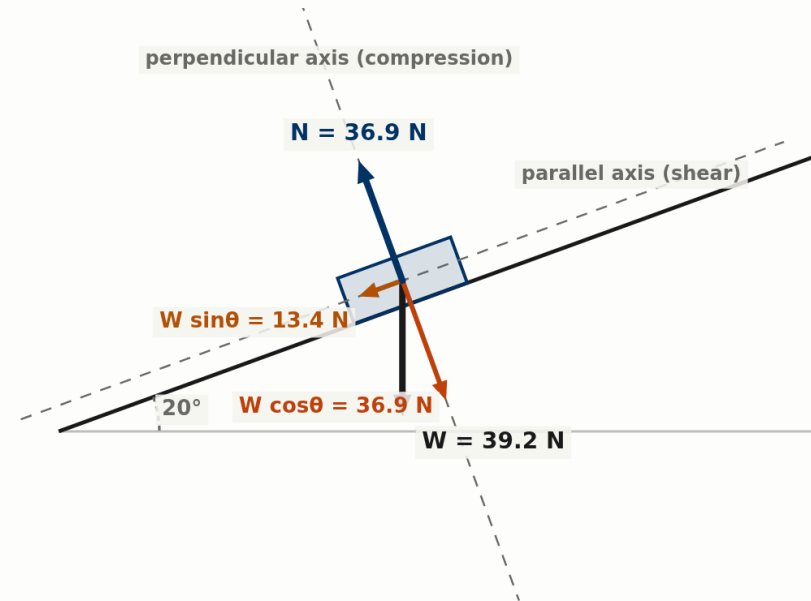
Mass 4 kg

flat ground 4 kg at 20° a 102 kg cyclist at 20°

What would happen to:

- The force of gravity
- Normal force
- and force components

If we change the angle of the contact surface?



At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

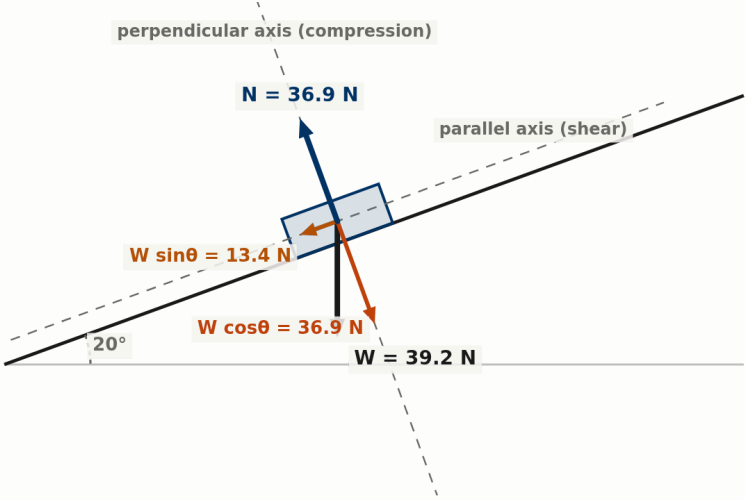
Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle 20°

Mass 4 kg

flat ground 4 kg at 20° a 102 kg cyclist at 20°

How to resolve for the components of force on a slant surface and solve for the normal force. We need to know the angle of the slant (20°) and the mass of the object (4 kg).



4.0 kg on a 20° slope

the weight
 $W = mg = 4.0 \times 9.81 = 39.24\text{ N}$
perpendicular — pressed into the surface
 $W \cos\theta = 39.24 \cdot \cos 20^\circ = 36.87\text{ N}$
parallel — sliding down the surface
 $W \sin\theta = 39.24 \cdot \sin 20^\circ = 13.42\text{ N}$

N	$shear$	N / W
36.9 N	13.4 N	94%

At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope
Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle

Mass

20°

4 kg

flat ground

4 kg at 20°

a 102 kg cyclist at 20°

4th — solve for the parallel and perpendicular components of force.

Parallel component

$$\text{Opposite} = H \cdot \sin 20^\circ$$

$$\text{Opposite} = 39.24 \cdot \sin 20^\circ$$

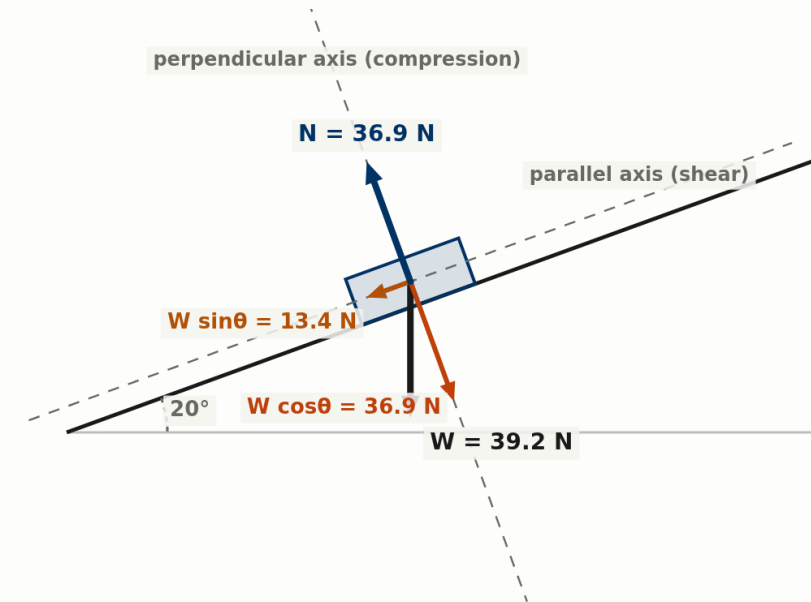
$$\text{Opposite} = 13.42 \text{ N}$$

Perpendicular component

$$\text{adjacent} = H \cdot \cos 20^\circ$$

$$\text{adjacent} = 39.24 \cdot \cos 20^\circ$$

$$\text{adjacent} = 36.87 \text{ N}$$



At 20° the normal force is **36.9 N**, not 39.2 N — the missing 2.4 N has become **13.4 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle 20°

Mass 4 kg

flat ground 4 kg at 20° a 102 kg cyclist at 20°

Force

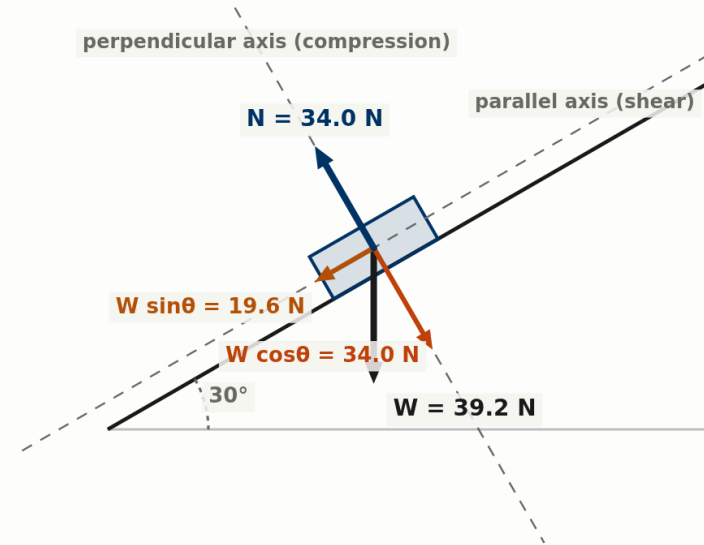
In general there is a simple rule to determine the parallel and perpendicular components of force.

For any incline of a certain angle the parallel and perpendicular components of force will be:

$$\text{parallel} = mg \sin \theta$$

$$\text{perpendicular} = mg \cos \theta$$

Where mg is the equation for the weight of the object.



At 30° the normal force is **34.0 N**, not 39.2 N — the missing 5.3 N has become **19.6 N of shear** down the slope

Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens.

Slope angle 30°

Mass 4 kg

flat ground 4 kg at 20° a 102 kg cyclist at 20°

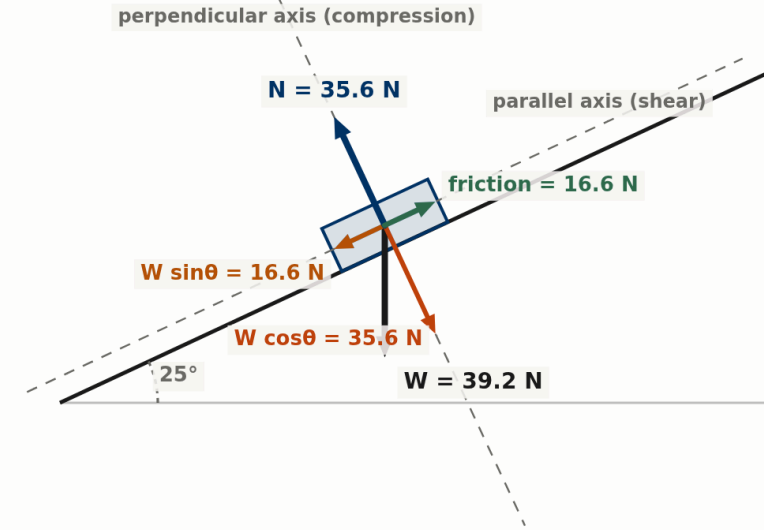
Friction

The parallel component of gravity will tend to pull the block down the surface...

Friction force is required to be equal and opposite of the parallel force or the block would slide down the slope.

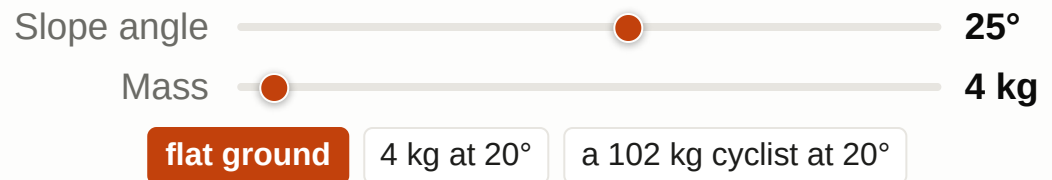
Friction force is proportional to the normal force and acts perpendicular to it.

We will discuss more about friction in a following lecture.



At 25° the normal force is **35.6 N**, not 39.2 N — the missing 3.7 N has become **16.6 N of shear** down the slope

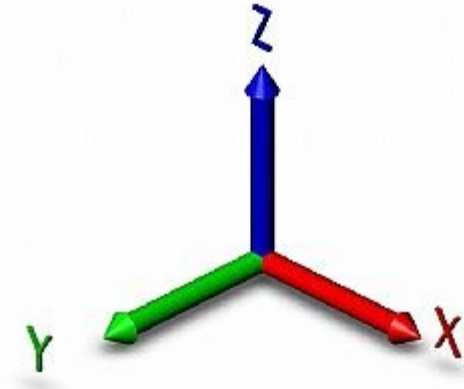
Gravity never changes — it is the same arrow straight down at every angle. What changes is the axis system we measure it against, which is why the normal force falls away as the slope steepens, and why friction has to hold up whatever the shear is.



Forces in 3D

When we move we exert force in **all directions**.

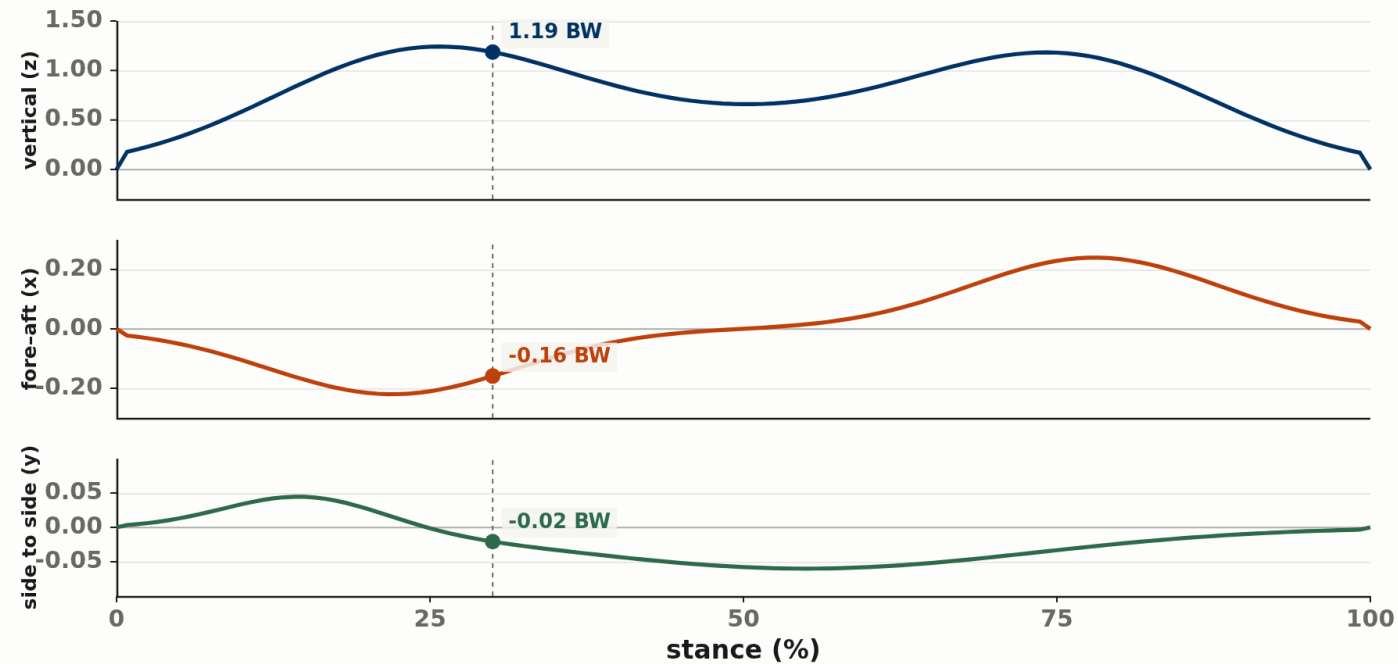
Both internal and external forces are resolved into components in reference to anatomical planes and axis systems.



Force in 3D

When considered in a **sagittal plane**: when we start to walk we exert a force that is not just acting down (z-axis) — part of the force is parallel to the surface (x-axis).

Also when considered in a **frontal plane**: part of the force is parallel to the surface acting along the y-axis.



30% of stance · vertical **1.19** · fore-aft **-0.16** · side to side **-0.02** BW

The fore-aft component is **negative**: the ground is pushing backwards on you. This is the braking half of the step. The vertical component is twenty times the others, which is why it is the one everybody quotes — and why the small ones are where the interesting things hide.

Stance

30%

▶ Take a step



Measuring external force in biomechanics

- Strain gauges
- Force Transducers
- Force plates
- Load cells
- Springs
- Piezoelectric crystals
- Dynamometer

An applied force will cause an object to deform and this deformation is measured and amplified by the electronics.



Measuring external force in biomechanics

In many clinical settings force transducers are simple **single axis** devices.

A good example is a hand-held dynamometer. This device is used to determine:

- muscle strength capabilities
- weaknesses



Measuring external force in biomechanics

Gold standard measurement of force in biomechanics is the **forceplate** and **instrumented treadmill**.

PRO'S

Accurate measure of ground reaction force

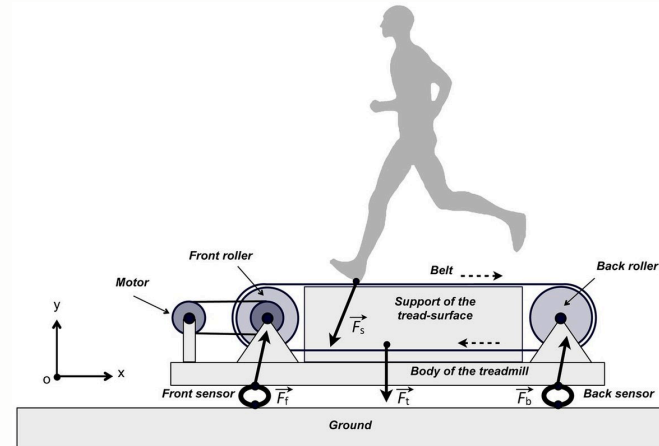
Reliable

Multidimensional

CON'S

Difficult to use in the field

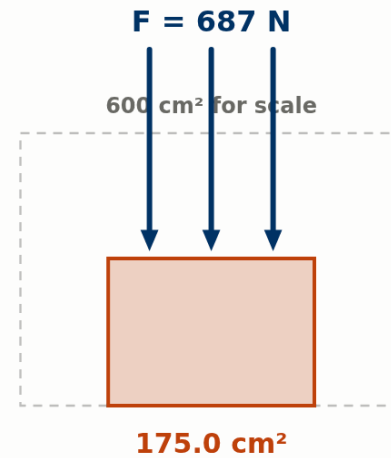
Cannot measure foot pressure distribution



Measuring external force in biomechanics

Often it is not feasible to directly measure external force but an inexpensive alternate is to measure **pressure**.

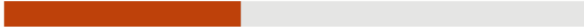
Pressure is the force per unit area directed perpendicular to an object surface: $P = F/A$



$$P = F / A$$
$$687 \text{ N} / 0.0175 \text{ m}^2$$

39.2 kPa

one foot flat


log scale — the spike is off the end of a linear one

687 N on **175.0 cm²** = **39.2 kPa**

The force never changed — only the area it is spread over. That is why a pressure sensor is not a force sensor: two readings can differ by a hundred times with the same load on the foot.

Body mass

70 kg

standing, both feet

one foot flat

heel strike

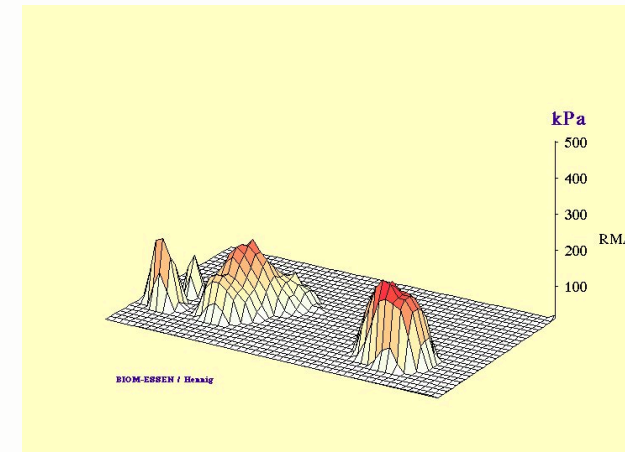
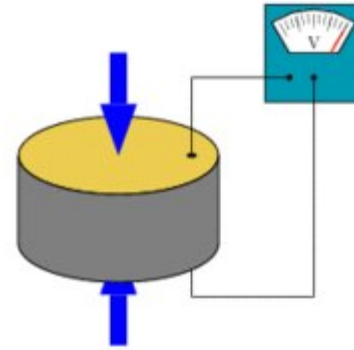
one running spike

both skis

How pressure sensors work

Most common pressure sensors measure pressure using the **piezoresistive effect**.

- **Piezo** – means to squeeze or press
- The piezoresistive effect is a change in the electrical resistivity of a semiconductor or metal when mechanical strain is applied.



Measuring external force in biomechanics

A common research and field based measure of foot loading is **pressure sensing technology**.

PRO'S

Portable, thin, flexible, lightweight

Approximation of force

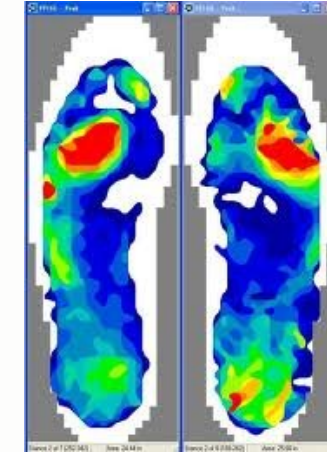
Accurate for timing of foot loading

Can measure foot pressure distribution while force plates cannot

CON'S

Unidimensional (vertical loading)

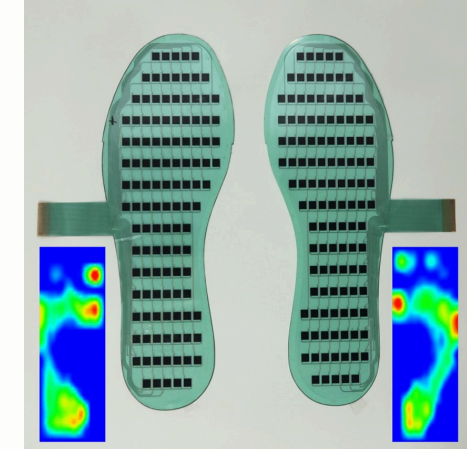
Degree of precision, accuracy and reliability requires testing – refinement



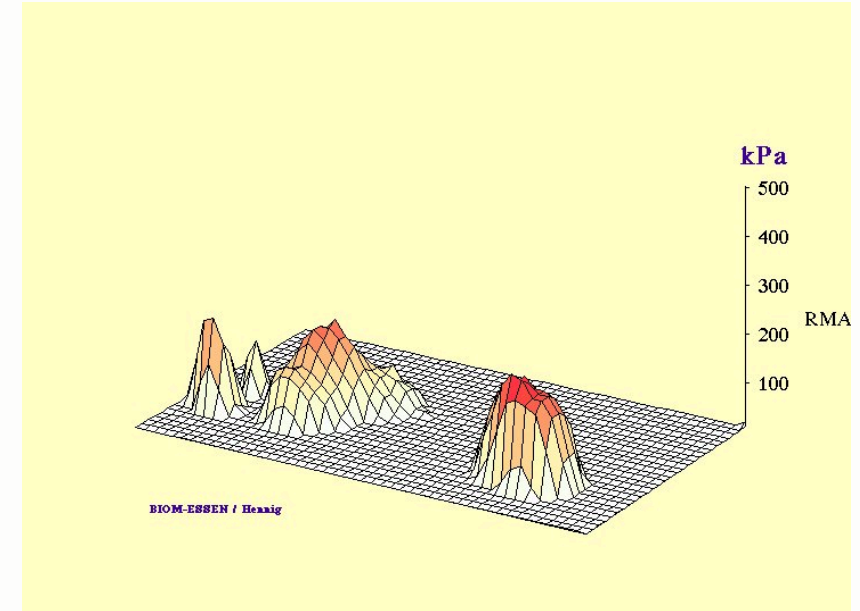
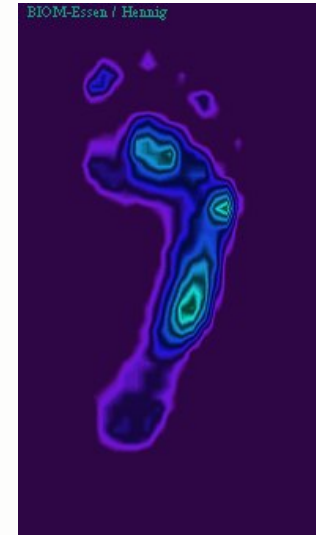
Measuring external force in biomechanics

There are many insole **plantar pressure measurement systems (PPMS)**.

These sensors help to provide the potential to measure external pressure (and calculate external force).

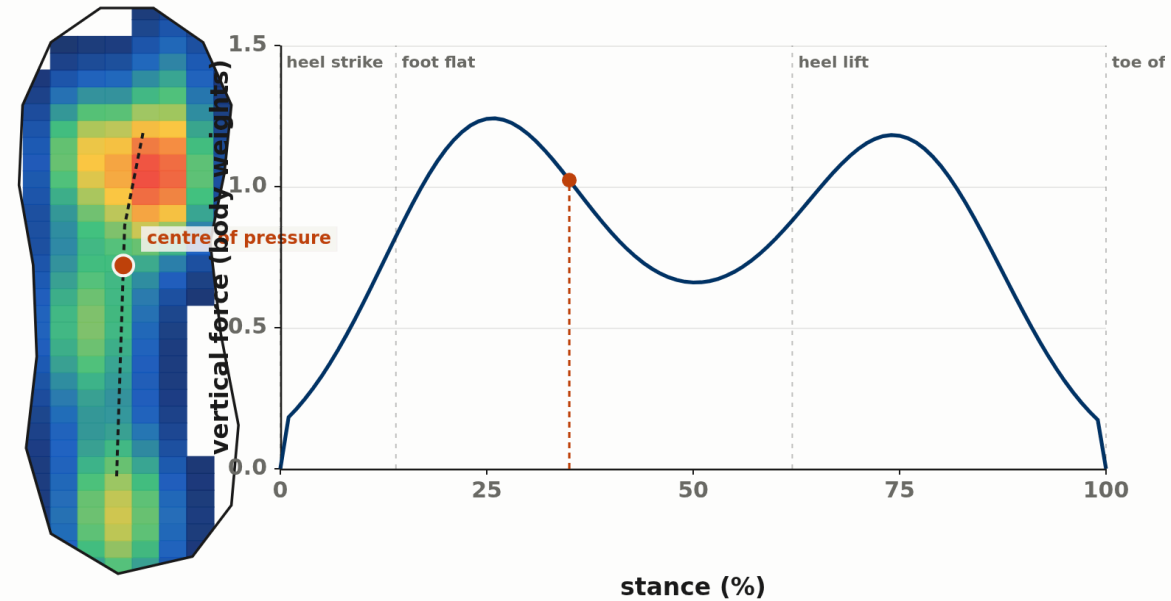


Plantar pressure sensors



Centre of pressure

Centre of Pressure is the **instantaneous point of application** of the ground reaction force. Measuring foot centre of pressure movement during walking can tell us more information about the GRF vector and how we use our feet while walking.



35% of stance · rolling forward through the midfoot · vertical force **1.02** body weights

The centre of pressure is the single point the ground reaction force acts through at that instant — not where the pressure is highest, but the average of the whole field. It sweeps heel to toe in about 0.6 s of walking.

Stance



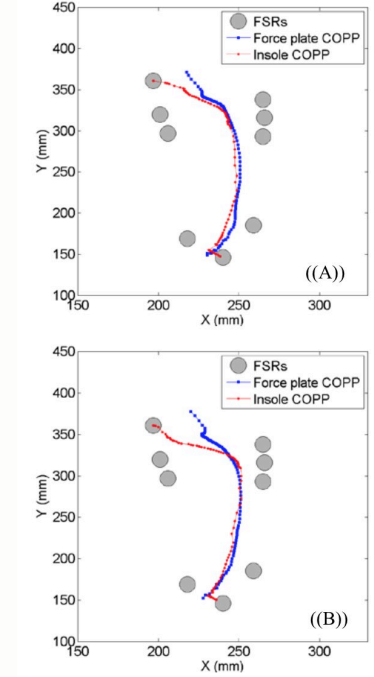
35%

▶ Take a step

Hide the path

Centre of pressure

PPMS can tell us similar information about COP as compared to force plates.



Measuring external force in biomechanics

Instrumented devices

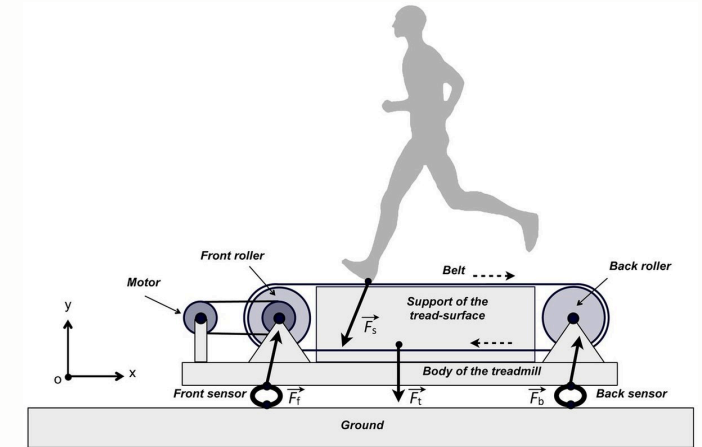
Force and pressure sensors can be added to mobility devices so that these objects become “instrumented” and we can measure the forces that individuals produce during activity.



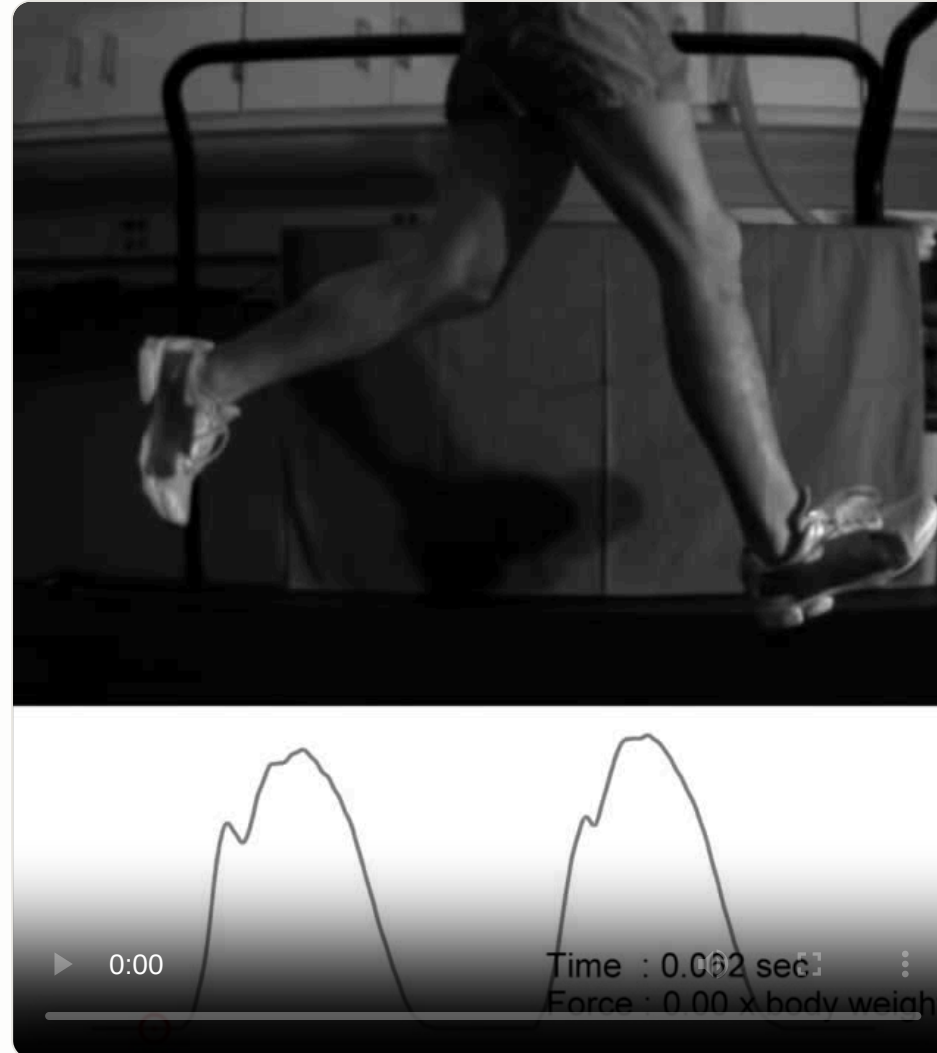
Force plate application

- Running style
 - Heel strike
 - Forefoot strike
- Footwear
 - Barefoot
 - Shoes
 - Vibram five fingers

Foot strike patterns and collision forces in habitually barefoot versus shod runners. Lieberman et al. *Nature*
nature.com/nature/journal/v463/n7280/pdf/nature08723.pdf

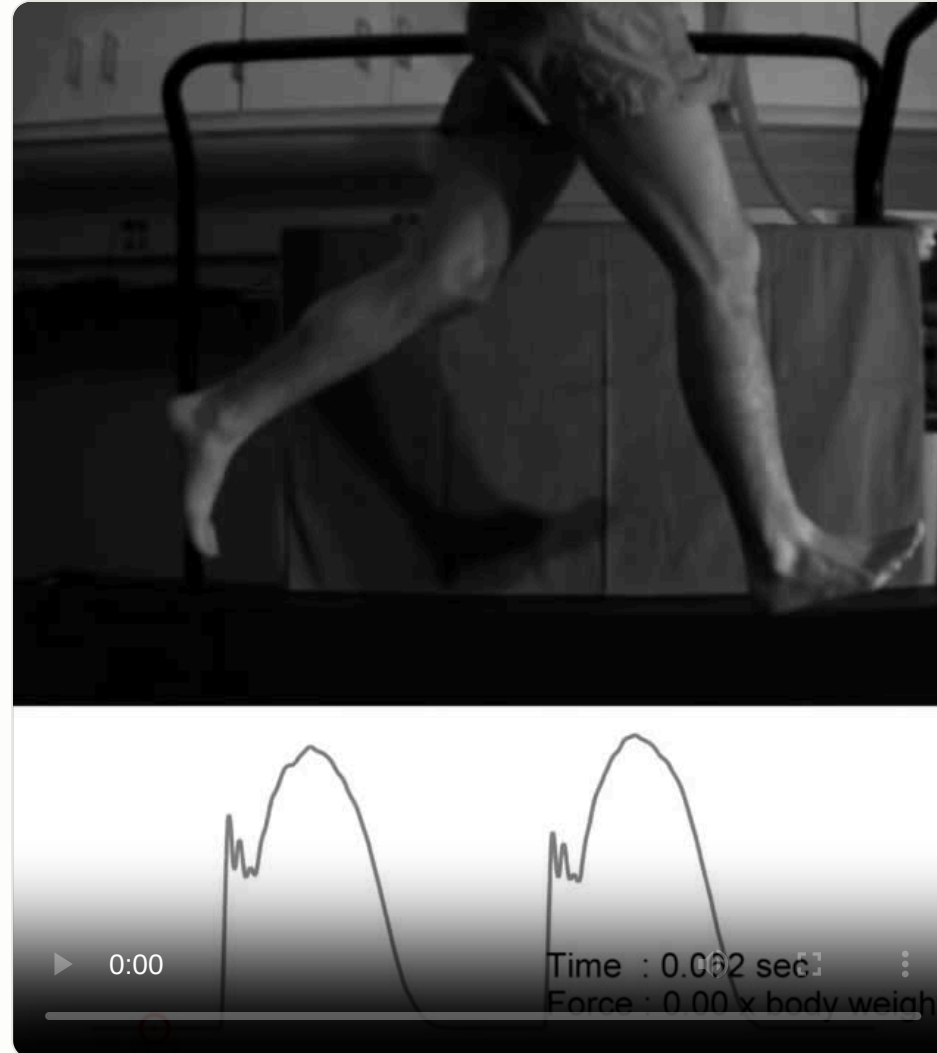


Shod heel strike



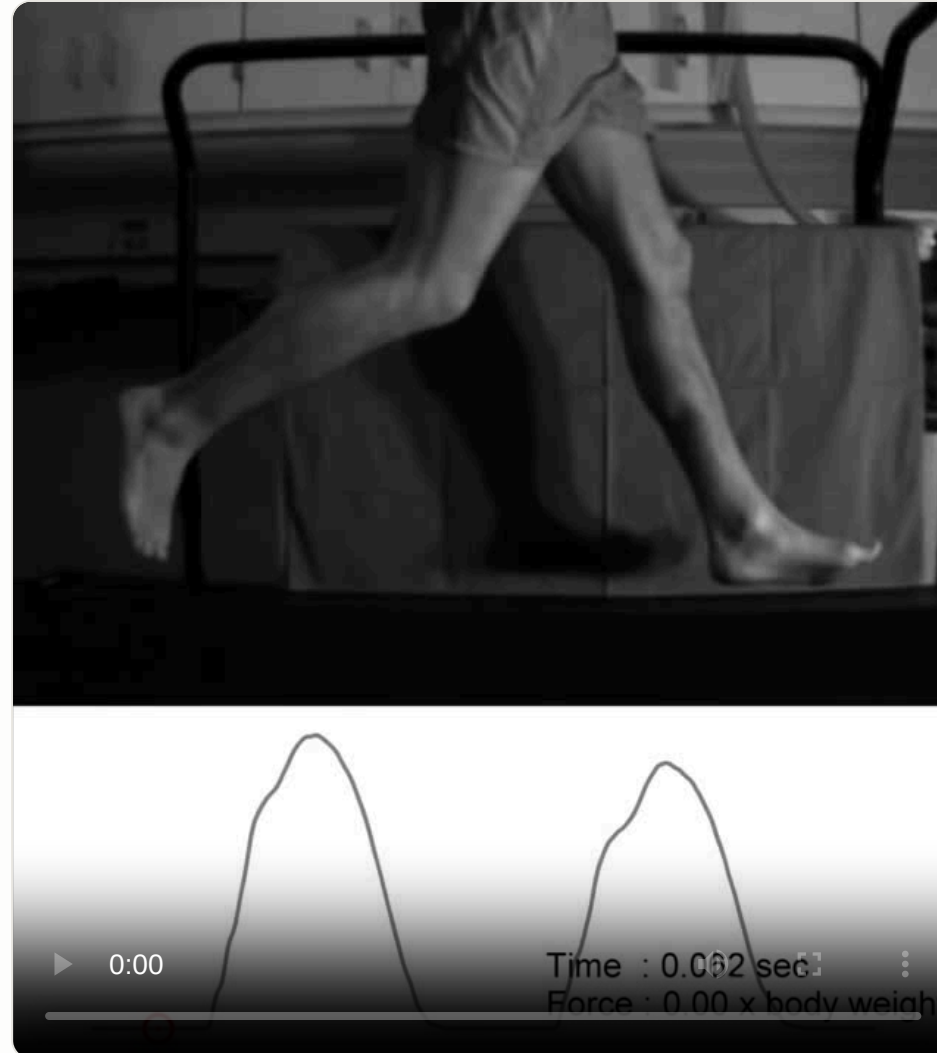
Slow motion with the vertical ground reaction force underneath.

Barefoot heel strike



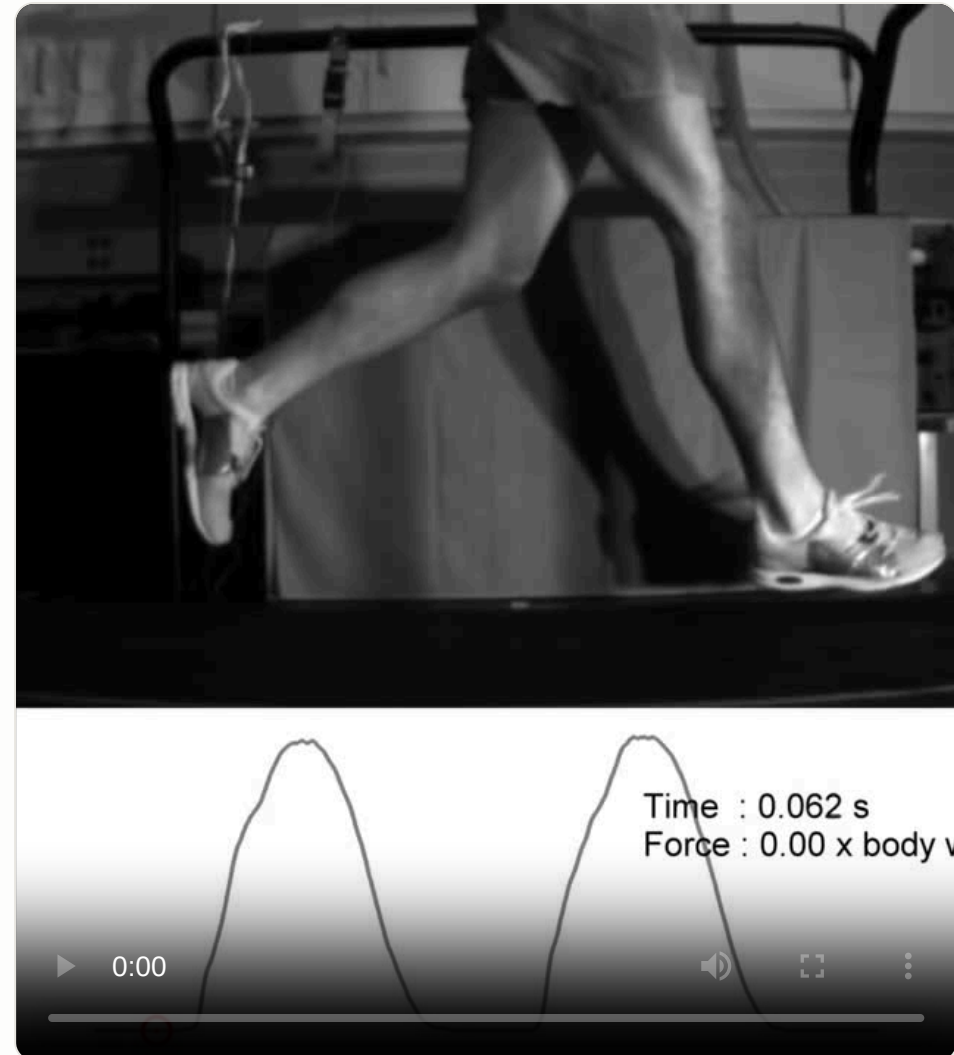
Same runner, same heel strike, no shoe.

Barefoot forefoot strike



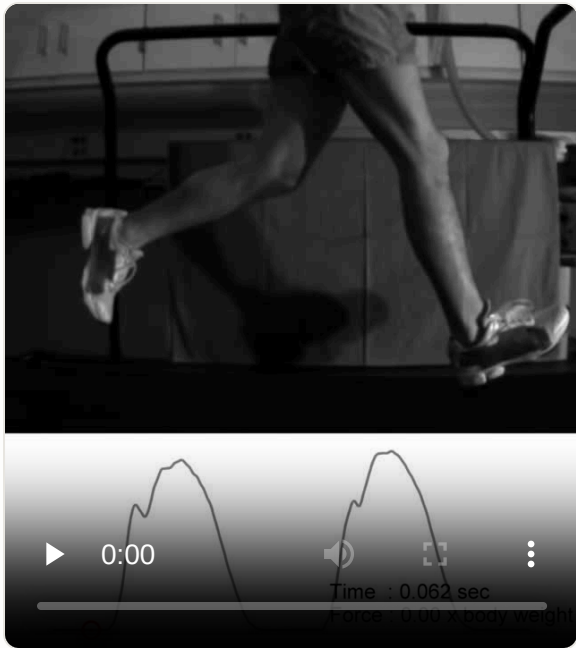
The forefoot lands first and the heel comes down afterwards.

Racing flat, forefoot strike



A forefoot strike in a thin-soled racing flat.

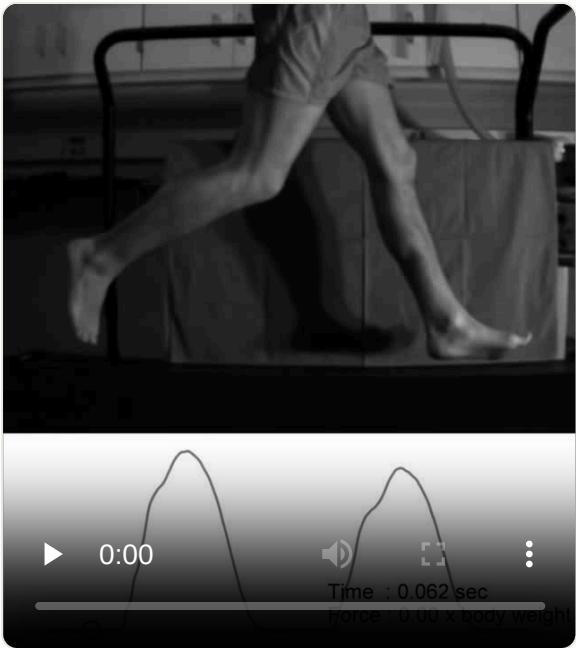
All four together



shod, heel strike



barefoot, heel strike



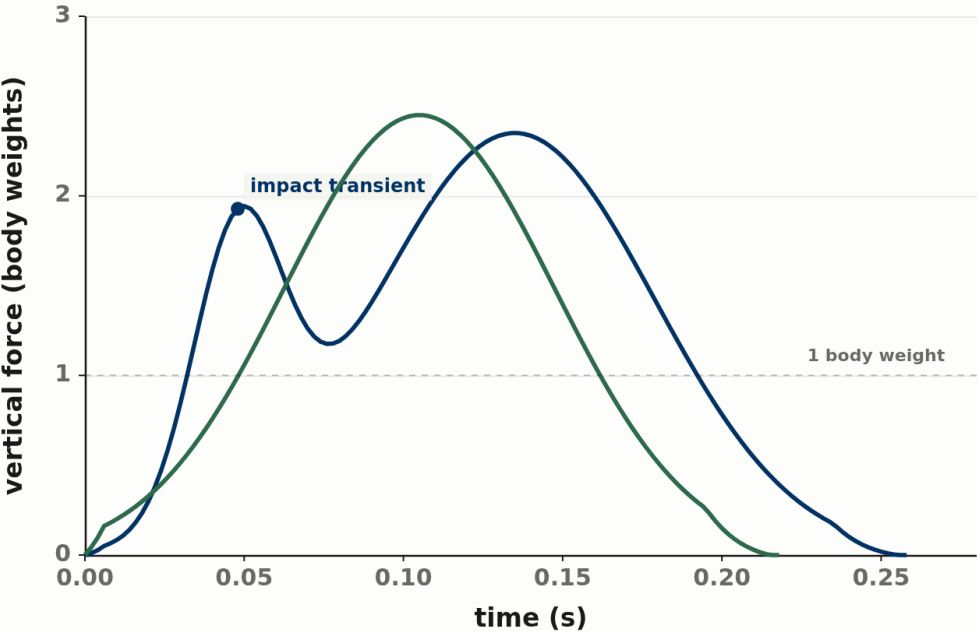
barefoot, forefoot strike



racing flat, forefoot strike

The same comparison, live

Every style lands on about the same peak force. What separates them is the spike in the first 50 milliseconds — and how fast the load arrives.



peak force and how fast it arrives

	peak	in N	rate
shod, heel strike	2.35	1614	75
barefoot, heel strike	2.71	1860	264
barefoot, forefoot	2.45	1682	35

Peak in body weights, the same force in newtons for a 70 kg runner, and the steepest rise in body weights per second.

Every style ends up near the same peak. What separates them is the **spike in the first 50 ms** and how quickly the load arrives — which is the part a heel strike in a cushioned shoe softens, and a bare heel does not.

shod, heel strike 2.35 BW · barefoot, forefoot 2.45 BW

Modelled on Lieberman et al., Nature 2010, who measured this on an instrumented treadmill — peak, timing and loading rate follow the published curves.

shod, heel strike

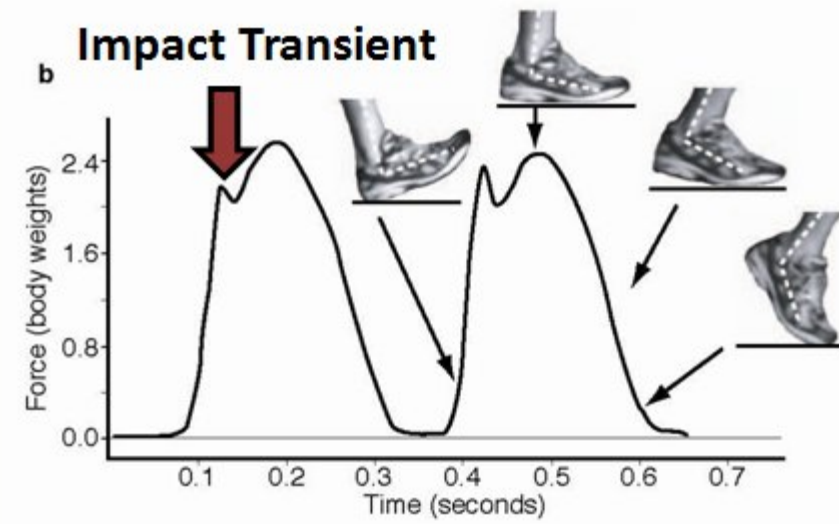
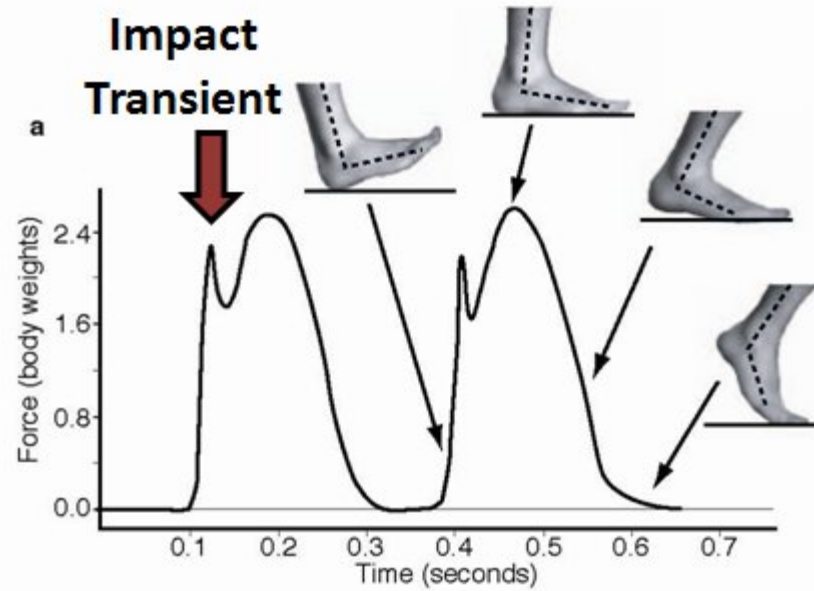
barefoot, heel strike

barefoot, forefoot

Runner mass

70 kg

The published traces

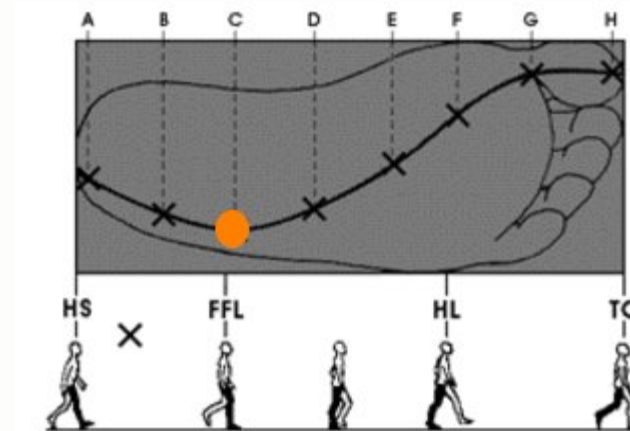
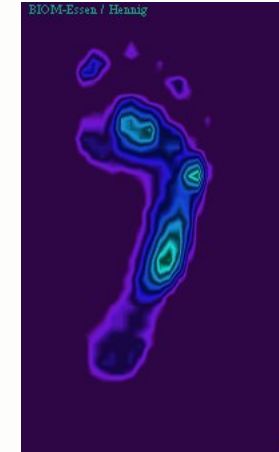
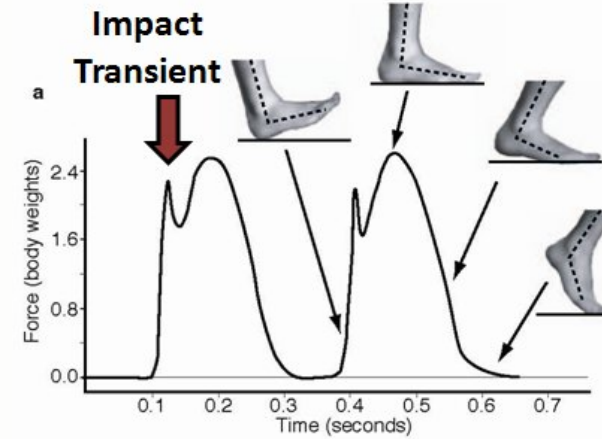


Lieberman et al., *Nature* 463, 531–535 (2010). The arrow marks the impact transient.

Measuring external force in biomechanics

Daniel Lieberman

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Measuring internal forces in Biomechanics

While force plates and pressure sensors can be used to measure external forces, it is difficult to measure the forces at the bones, joints and tendons directly.

But we can use different methods to approximate the internal forces:

- inverse dynamics
- electromyography

